Geonwoo Cho

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Research Summary

I study how reinforcement learning agents discover reusable skills and make decisions across long horizons. My current work asks when contrastive learning yields representations that generalize across goals, how exploration can uncover skills that unlock later behaviors, and how to assign credit for delayed outcomes. My previous work developed methods for unsupervised skill discovery, curriculum design, and stable adaptation from offline data to online interaction.

Education

University of Michigan — *Ann Arbor, MI*
PH.D. STUDENT IN COMPUTER SCIENCE AND ENGINEERING. ADVISOR: SILVIU PITIS. — *Aug 2026–present*

Gwangju Institute of Science and Technology (GIST) — *Gwangju, South Korea*
B.S. IN ELECTRICAL ENGINEERING AND COMPUTER SCIENCE, *CUM LAUDE*. MINOR IN MATHEMATICS. — *Feb 2019–Aug 2026*
Leave of absence for mandatory military service, Mar 2022–Feb 2024.

University of California, Berkeley — *Berkeley, CA*
EXCHANGE STUDENT, FUNDED BY GIST — *Jan 2025–Aug 2025*

Korea Science Academy of KAIST — *Busan, South Korea*
HIGH SCHOOL DIPLOMA — *Mar 2016–Feb 2019*

Publications

INTERNATIONAL CONFERENCES

- **Geonwoo Cho**, Jaegyun Im, Jihwan Lee, Hojun Yi, Sejin Kim, Sundong Kim. TRACED: Transition-Aware Regret Approximation with Co-Learnability for Environment Design. **ICLR 2026**.
- **Geonwoo Cho**[*], Jaemoon Lee[*], Jaegyun Im, Subi Lee, Jihwan Lee, Sundong Kim. AMPED: Adaptive Multi-Objective Projection for Balancing Exploration and Skill Diversification. **ICLR 2026**.

WORKSHOPS

- **Geonwoo Cho**, Jaegyun Im, Doyoon Kim, Lexin Li. ANCHOR: KL-Anchored Adaptation for Stable Offline-to-Online RL. CoRL 2026 Workshop on Continually Self-Improving Robots.
- Jihwan Lee, **Geonwoo Cho**, Hojun Yi, Jaegyun Im, Giseung Park, Sundong Kim. Receding-Horizon Execution for Action Chunking in Offline-to-Online Reinforcement Learning. ICML 2026 Workshop on Decision-Making from Offline Datasets to Online Adaptation.
- **Geonwoo Cho**, Jaegyun Im, Doyoon Kim, Sundong Kim. Causal-Paced Deep Reinforcement Learning. RLC 2025 Workshop on Causal Reinforcement Learning (**oral**).

KOREAN CONFERENCES

- **Geonwoo Cho**[*], Subi Lee[*], Jaemoon Lee. Evaluating Simplicial Normalization in Multi-Task Reinforcement Learning. Korea Software Congress, 2024 (poster).
- Seung-Je Cho, **Geonwoo Cho**, Young-wook Kim. Development of a Deep Learning-Based House-Tree-Person Test Analysis Model. Korea Information Processing Society Fall Conference, 2021 (poster).
- **Geonwoo Cho**, Dong Keon Park, Hong Kook Kim. LSTM-Based Earthquake Anomaly Detection Applied to Total Electron Current Data. Korea Artificial Intelligence Conference, 2020 (poster).

* Equal contribution.

Honors & Awards

Aug 2026 **Future Talent Award**, GIST
Aug 2026 **Outstanding Thesis Award**, GIST
Fall 2024 **Academic Excellence Scholarship**, GIST
Dec 2020 **Dream AI Open Challenge**, Fourth Prize, Korea Ministry of Science and ICT
Jun 2020 **International Collegiate Programming Contest (ICPC)**, Seoul Regional qualifier
Dec 2019 **GIST President's Award**, First Prize, Creative Convergence Competition

Research Experience

University of Michigan *Ann Arbor, MI*
DOCTORAL RESEARCH. ADVISOR: SILVIU PITIS. *Aug 2026–present*
Analyze the spectral structure of goal-reaching representations in contrastive RL. Develop Empowered Skill Trees to direct exploration toward skills that unlock later behaviors.

University of California, Los Angeles
STATISTICS AND DATA SCIENCE. ADVISOR: YUHUA ZHU. *Jun 2025–Aug 2026*
Investigated a partial differential equation formulation for continuous-time offline reinforcement learning.

University of California, Berkeley
BIOSTATISTICS. ADVISOR: LEXIN LI. *Jan 2025–Aug 2026*
Developed ANCHOR to stabilize online fine-tuning of a pretrained policy when its offline training data are unavailable.

GIST, Data Science Lab
ADVISOR: SUNDONG KIM *Apr 2024–Aug 2026*
Developed methods for unsupervised skill discovery and curriculum design, including AMPED, TRACED, and Causal-Paced Deep Reinforcement Learning.

GIST, AITER Lab
ADVISOR: HONG KOOK KIM *Jun 2020–Dec 2020*
Applied time-series models to ionospheric measurements for earthquake anomaly detection.

Industry Experience

Team Learners
MACHINE LEARNING SOFTWARE ENGINEER *Aug 2023–Jan 2024*
Reduced diffusion-model inference time through computation graph optimization.

Hyperconnect (Match Group)
MACHINE LEARNING SOFTWARE ENGINEER *Jun 2022–Jul 2023*

- Built a transformer-based matchmaking system serving 1,000 requests per second with less than 0.001% downtime. The model increased revenue by 3 percentage points and retention by 7 percentage points.
- Reduced feature-store p99 latency from 200 ms to 150 ms by changing database access patterns and adopting Avro serialization.

Business Canvas
SOFTWARE ENGINEER *Dec 2021–Jun 2022*
Improved service availability to 99.95% through microservice architecture and better observability.

Algorima
SOFTWARE ENGINEER *Dec 2020–Jun 2021*
Built web services and machine learning pipelines.

Patent Applications

- Method and System for Gradient Surgery-Based Multi-Objective Reinforcement Learning. No. KR 10-2025-0200580 (Dec 2025).
- Method and System for Task Prioritization Reinforcement Learning Based on Structural Differences. No. KR 10-2025-0196565 (Dec 2025).
- Methods and Systems for Learning Based on Difficulty and Mutual Learning. No. KR 10-2025-0160737 (Oct 2025).

Selected Talks

- Causal-Paced Deep Reinforcement Learning. Contributed oral, RLC Workshop on Causal Reinforcement Learning, Aug 2025.
- Feature Store Implementation for Real-Time Recommender Systems. Dev Night, GIST, Sep 2024.

Teaching & Professional Service

Industry Mentor, F-Lab. Software Engineering and DevOps. *Feb 2024–Jan 2025*
Teaching Assistant, GIST. Single Variable Calculus and Applications, Machine Learning and Deep Learning. *Feb–Dec 2024*
Reviewer, ICLR 2027, NeurIPS 2026, and the CoRL 2026 Workshop on Continually Self-Improving Robots.